



HW 13

65070021 kittipon



Quiz

- จงแสดงขั้นตอนการหาแก้ปัญหาดังนี้ โดยใช้แนวคิดแบบ Divide and Conquer
 - $2^{1000} \bmod 10$
 - $3^{600} \bmod 5$
- Binary Search จัดเป็น Divide and conquer algorithm หรือไม่ เพราะเหตุใด
- จงกำหนด base case และ recursive case ของอัลกอริทึม binary search

① $2^{1000} \bmod 10 = (2^{500} \bmod 10)^2 \bmod 10 = \textcircled{6} \neq$

$2^{500} \bmod 10 = (2^{250} \bmod 10)^2 \bmod 10 = \textcircled{6}$

$2^{250} \bmod 10 = (2^{125} \bmod 10)^2 \bmod 10 = \textcircled{4}$

$2^{125} \bmod 10 = 2 \cdot (2^{62} \bmod 10)^2 \bmod 10 = \textcircled{2}$

$2^{62} \bmod 10 = (2^{31} \bmod 10)^2 \bmod 10 = \textcircled{4}$

$2^{31} \bmod 10 = 2 \cdot (2^{15} \bmod 10)^2 \bmod 10 = \textcircled{8}$

$2^{15} \bmod 10 = 2 \cdot (2^7 \bmod 10)^2 \bmod 10 = \textcircled{8}$

$2^7 \bmod 10 = 2 \cdot (2^3 \bmod 10)^2 \bmod 10 = \textcircled{8}$

$2^3 \bmod 10 = 2 \cdot (2^1 \bmod 10)^2 \bmod 10 = \textcircled{8}$

$2^1 \bmod 10 = 2 \cdot (2^0 \bmod 10)^2 \bmod 10 = \textcircled{2}$

$2^0 \bmod 10 = \textcircled{1}$

②

$$3^{600} \bmod 5 = (3^{300} \bmod 5)^2 \bmod 5 = 1$$

$$3^{300} \bmod 5 = (3^{150} \bmod 5)^2 \bmod 5 = 1$$

$$3^{150} \bmod 5 = (3^{75} \bmod 5)^2 \bmod 5 = 4$$

$$3^{75} \bmod 5 = 3 \cdot (3^{37} \bmod 5)^2 \bmod 5 = 2$$

$$3^{37} \bmod 5 = 3 \cdot (3^{18} \bmod 5)^2 \bmod 5 = 3$$

$$3^{18} \bmod 5 = (3^9 \bmod 5)^2 \bmod 5 = 4$$

$$3^9 \bmod 5 = 3 \cdot (3^4 \bmod 5)^2 \bmod 5 = 3$$

$$3^4 \bmod 5 = (3^2 \bmod 5)^2 \bmod 5 = 1$$

$$3^2 \bmod 5 = (3^1 \bmod 5)^2 \bmod 5 = 4$$

$$3^1 \bmod 5 = 3 \cdot (3^0 \bmod 5)^2 \bmod 5 = 3$$

$$3^0 \bmod 5 = 1$$

